

# **ALY 6060 Decision Support and Business Intelligence**

## **Module 5 – Case Study**

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## **A. Summary**

We analyzed a dashboard focused on identifying patterns related to errors, debt, income, and transaction types for PayPal to address the question: Which factors most significantly predict fraud, and how can this insight minimize fraudulent transactions? Online transactions have higher error rates, indicating greater fraud susceptibility. Retirement age groups (65–70) are more prone to errors, suggesting potential targeting by fraudsters. Cities with higher error rates could serve as fraud hotspots. Debt exceeding income highlights financially vulnerable users who may be more susceptible to fraud.

Recommended strategies for fraud mitigation include strengthening authentication for online transactions, tailoring fraud education for vulnerable demographics, implementing real-time fraud detection systems, and analyzing regional patterns to target high-risk areas. These insights could guide predictive modeling and adaptive fraud prevention systems to minimize fraudulent activity effectively.

## **B. Findings from the Case Study**

### **1. Enhanced Fraud Detection Accuracy:**

Using H2O Driverless AI, PayPal achieved a dramatic improvement in fraud detection accuracy, increasing it from 0.89 to 0.947—a nearly 6% increase. This improvement surpassed the results of over 10 years of expert feature engineering. “It doesn’t have to stop with your own data. Shared intelligence could make machine learning algorithms for payment fraud detection even stronger. For instance, PayPal’s two-sided network is a rich source of transaction and risk data from millions of active global accounts that may help enhance fraud detection.” (PayPal Payment Fraud Detection, November 2024)

### **2. Discovery of Significant Features:**

The top five features identified by Driverless AI outperformed those previously engineered manually, showcasing the effectiveness of automated feature engineering.

### **3. Speed Improvements:**

Leveraging an IBM Power GPU-based server, model training became six times faster compared to CPU environments. This speed increase enhances PayPal’s ability to respond to emerging fraud patterns in real-time.

### **4. Novel Insights from Graph Network Analysis:**

By using graph databases (e.g., Neo4j) and algorithms like Node2Vec, PayPal identified relationships among buyers and sellers. This approach allowed them to detect fraudulent accounts by analyzing shared attributes such as IP addresses or shipping addresses within interconnected networks.

## **5. Scalable Solution:**

PayPal successfully processed a massive graph of 1.5 billion edges and 500,000 nodes, demonstrating the scalability of their solution to analyze large datasets effectively.

## **C. PayPal's Implementation of AI**

PayPal leverages AI and machine learning to enhance transaction authorization rates and combat fraud effectively. By predicting issuer declines, implementing smart retry strategies, and detecting fraud in real time, PayPal has achieved significant improvements. Key AI applications include:

### **1. Authorization Optimization:**

Machine learning models, such as Gradient Boosting Machines (GBM), analyze issuer decline patterns and recommend remedies, improving authorization rates by up to 240 basis points for some merchants and 30 bps through retry strategies alone. “PayPal uses machine learning models to improve the authorization rates of valid transactions on its platform by predicting and addressing the issuer declines that interrupt user payment requests.”(Emerge AI Research, Jan 2022)

### **2. Fraud Detection:**

Using graph technologies and real-time databases, PayPal analyzes complex transactional relationships, identifying fraud patterns across its extensive dataset from over 15 billion annual transactions.” To combat payment fraud, PayPal designed PayPal Fraud Protection, an “adaptive machine learning solution that helps merchants protect against evolving fraud.” PayPal claims that the strength of the solution in detecting fraud comes from the company’s vast stores of data, generated from its 2-Sided Network, the name PayPal uses to indicate the two sides of each transaction.”(Emerge AI Research, Jan 2022)

### **3. Results:**

These AI-driven solutions have reduced false declines, minimized fraud-related losses, and improved customer experiences. Between 2018 and 2020, transaction losses decreased from 0.18% to 0.12% of total payment value, significantly outperforming industry benchmarks.

AI continues to drive PayPal’s growth by improving efficiency, reducing costs, and fostering trust in its payment ecosystem.

## **D. Impact of AutoML on Results**

### **1. Automation of Feature Engineering:**

H2O Driverless AI's automatic feature engineering uncovered new predictive factors that were previously unrecognized, eliminating the need for extensive manual labor by data scientists.” In the world of cybersecurity, things happen fast. Machine learning algorithms run hundreds of thousands of queries in milliseconds and can often assess individual customer behaviors in real-time. This makes it possible to quickly differentiate legitimate customers from fraudulent ones, helping you quickly approve authentic transactions and create a seamless experience for trusted customers.“ (PayPal Payment Fraud Detection, November 2024)

### **2. Improved Efficiency:**

The integration of advanced AI reduced the time to train models, enabling quicker detection of fraud patterns and thus reducing potential financial losses.

### **3. Advanced Fraud Detection Techniques:**

AI allowed PayPal to move beyond individual buyer or seller behaviors, enabling the detection of collusive fraud by analyzing entire networks of accounts and their interactions.” With machine learning, you can process large datasets with multiple variables and quickly uncover correlations that might indicate more sophisticated fraud attempts. Not only do these models identify trends and patterns that human eyes might miss, they’re also able to continuously adapt to the ever-changing fraud landscape.“ (PayPal Payment Fraud Detection, November 2024)

## **E. Benefits of Data Analysis to PayPal**

### **1. Enhanced Fraud Prevention:**

By combining AI and graph analysis, PayPal can now predict and prevent fraud more effectively, reducing financial losses and protecting their customers.

### **2. Improved Customer Trust:**

Accurate fraud detection ensures that genuine customers benefit from PayPal’s buyer and seller protection programs, strengthening customer loyalty.

### **3. Actionable Insights into Fraud Networks:**

Data analysis helped uncover relationships within fraudulent networks, enabling PayPal to identify and preemptively shut down interconnected accounts before significant damage occurs.

### **4. Scalable and Adaptive Models:**

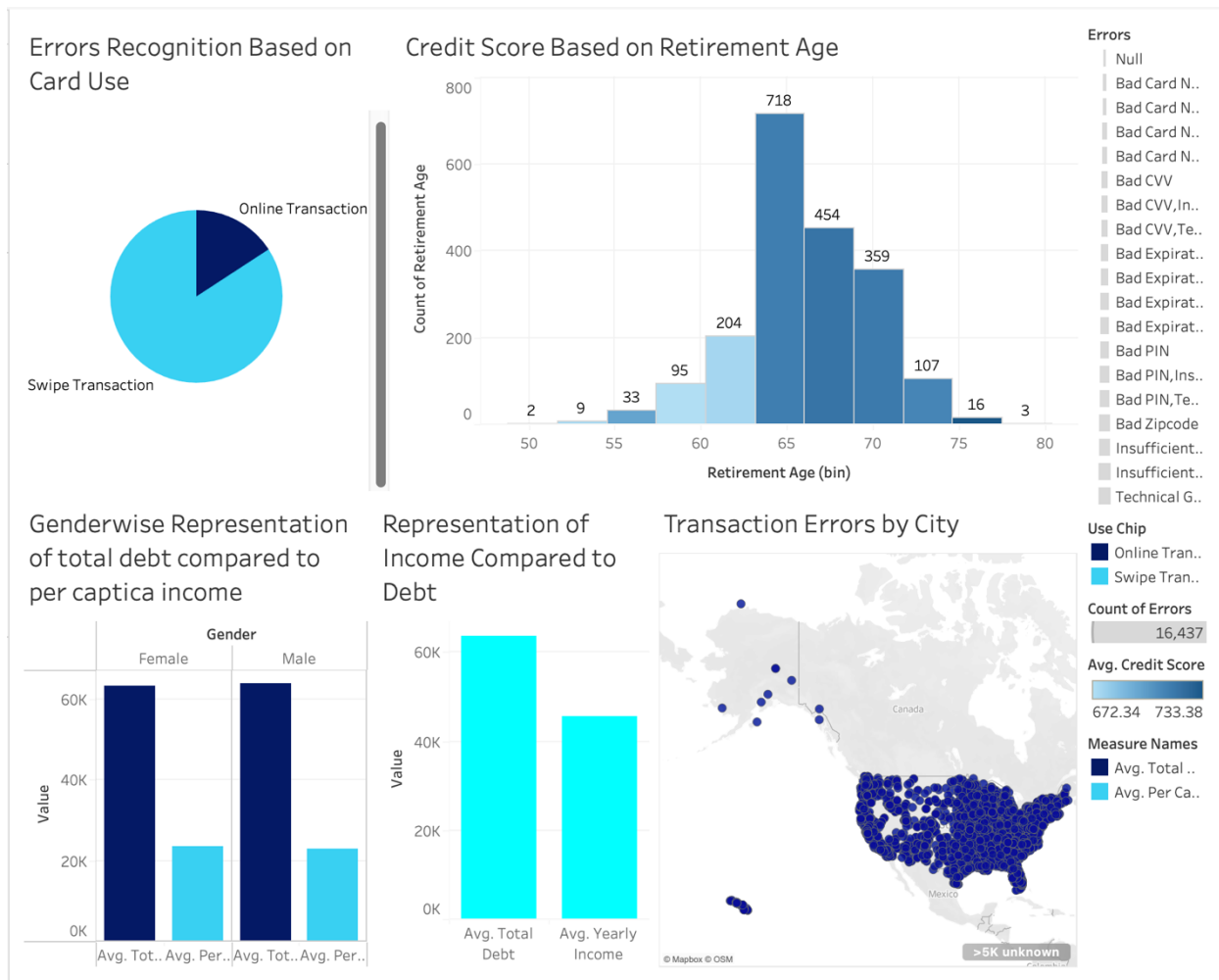
PayPal's ability to analyze vast datasets and adapt to emerging fraud tactics positions the company to stay ahead of evolving fraud strategies.

## 5. Operational Efficiency:

Faster model training and reduced dependency on manual feature engineering allow PayPal's teams to focus on strategic tasks, optimizing their resources.

From the PayPal case study, the below question would be considered for further analysis:

**"Which factors most significantly predict fraud and how can we use this insight to minimize fraudulent transactions?"**



## F. Key Observations from the above dashboard:

### 1. Errors Recognition Based on Card Use:

Online transactions have a larger share of errors compared to swipe transactions. Online transactions are more prone to fraud due to less stringent authentication compared to physical card usage.

**Actionable Insight:** Strengthen authentication mechanisms for online transactions, such as multi-factor authentication (MFA) or additional verifications like OTPs.

## **2. Credit Score Based on Retirement Age:**

Retirees around ages 65–70 appear to have high transaction counts and errors. Fraudsters may target older individuals, as they might be perceived as less tech-savvy or vigilant.

**Actionable Insight:** Implement tailored fraud alerts and educational campaigns for users in older age groups to recognize potential scams.

## **3. Gender-wise Representation of Debt and Income:**

Both genders have similar levels of debt compared to income, but financial stress may vary due to other factors not shown here. Individuals with higher debt-to-income ratios may be more susceptible to fraudulent schemes, like phishing for debt relief.

**Actionable Insight:** Monitor users with high debt-to-income ratios for unusual activity and introduce protective measures for financially stressed individuals.

## **4. Transaction Errors by City:**

Certain cities report more errors in transactions. Cities with higher error rates could be hotspots for fraudulent activities due to higher transaction volume or weaker detection systems.

**Actionable Insight:** Analyze city-specific error patterns to deploy fraud detection algorithms more effectively and focus resources on high-risk regions.

## **5. Representation of Income vs. Debt:**

Debt often exceeds annual income for many users. Financially stressed individuals may become targets for fraudsters offering fake financial relief services.

**Actionable Insight:** Provide financial education programs and monitor for fraudulent transactions targeting financially vulnerable users.

## **6. Data-Driven Fraud Prediction:**

Using **machine learning models** (e.g., decision trees, gradient boosting, or neural networks) to analyze patterns in the dataset, including transaction type, user demographics, and error rates, to predict fraudulent behavior.

Include variables such as transaction amount, location, time, user history, and error type in the models.

#### **7. Fraud Minimization Strategies:**

Introduce real-time fraud detection systems that analyze multiple factors (e.g., transaction type, location, user history) before authorizing transactions.

Implement adaptive fraud prevention measures, such as higher scrutiny for transactions involving flagged regions, demographics, or online payment methods.

#### **G. Conclusion**

AI has significantly enhanced PayPal's ability to detect and prevent fraud, not only improving accuracy and speed but also enabling the detection of complex fraud schemes. By leveraging advanced analytics and AI-driven insights, PayPal protects its users, maintains trust, and ensures the integrity of its platform. This case exemplifies the transformative potential of AI in addressing real-world challenges like fraud prevention at scale.

## References

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